De Mathematics Competitions

3rd Annual

DIME II

De Invitational Mathematics Examination II

Friday, September 25, 2026



INSTRUCTIONS

1. DO NOT TURN THE PAGE UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 15-question competition. All answers are integers ranging from 000 to 999, inclusive. Make sure each of your answers is 3 digits, even if the first digit is zero.
3. Mark your answer to each problem on the answer sheet by completely filling the circle with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet and appear sufficiently dark will be scored.
4. SCORING: You will receive 1 point for each correct answer, 0 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, compasses, and erasers are allowed as aids. Prohibited materials include calculators, smartwatches, phones, computing devices, protractors, and graph paper.
6. Figures are not necessarily drawn to scale.
7. You will have 3 hours to complete the competition once your competition manager tells you to begin, though all work must be submitted by Friday, January 29, by 11:59 pm ET regardless of starting time.
8. You may take the DIME as many times as you want. Taking both the DIME I and II is highly encouraged.

The problems and solutions for this DIME were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

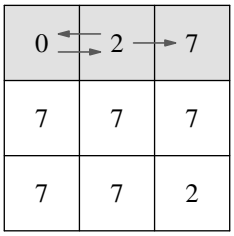
The DMC Committee reserves the right to disqualify scores if it determines that the rules were not followed. Participants are subject to the DMC Policies on Competition Integrity, which can be found on our site.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

Problem 1:
Michelle went to buy hats, jackets, and dresses at the fashion shop, where each hat, jacket, and dress costs \$20, \$50, and \$79, respectively. Suppose that Michelle spent \$1035 on the items in total and bought 25 items at the fashion shop, of which a were hats, b were jackets, and c were dresses. Find $a \cdot b \cdot c$.

Problem 2:
Convex quadrilateral $ABCD$ has $AB = 13$, $BC = 19$, and $\angle A$ and $\angle C$ right. Suppose that a point P lies inside $ABCD$ such that $\overline{AP} \parallel \overline{BC}$, $\overline{BP} \parallel \overline{CD}$, and $BP = 12$. Find the area of $ABCD$.

Problem 3:
Each cell of a 3×3 grid is to be labeled such that exactly one cell is labeled 0, exactly two cells are labeled 2, exactly six cells are labeled 7, and there exists a sequence of four (not necessarily distinct) cells, each sharing a side with the next, whose labels are 2, 0, 2, 7, in that order. One example is shown below.



Find the number of ways to label the cells.

Problem 4:
At a gathering of 10 people all with a shirt on, exactly 7 of them have a shirt on that matches that of at least one other person. Every pair of people whose shirts do not match have a conversation. Suppose that exactly 36 conversations occur. Find the number of subsets of 3 people in which no two have matching shirts on.

Problem 5:
Let a, b, and c be real numbers, and define the polynomials

$$P(x) = ax^2 + bx + c,$$
$$Q(x) = (a+1)x^2 + (b+11)x + (c+18),$$
$$R(x) = (a+2)x^2 + (b+37)x + (c+66).$$

The roots of P are p and q, the roots of Q are p and $2q$, and the roots of R are p and $4q$. Find $P(27)$.

Problem 6:
Let circles ω_1 and ω_2 have radii 22 and 5, respectively, and be internally tangent at P. Let a chord of ω_1 be tangent to ω_2 at Q with P and the center of ω_1 on opposite sides of the chord. Let line PQ intersect ω_1 at $R \neq P$. Suppose that the distance from the center of ω_1 to the chord is 3. Find QR^2.

Problem 7:
Find the number of (possibly empty) subsets S of $\{1, 2, 3, \ldots, 10\}$ such that each number in S can be colored tan or cocoa so that no two distinct same-colored numbers in S have greatest common divisor greater than 1.

Problem 8:
In acute $\triangle ABC$ with $AB = 16$ and $AC = 20$, let P and Q be the points on the line that bisects $\angle BAC$ such that $\angle ABP$ and $\angle ACQ$ are right, and let lines BP and CQ intersect at D. Suppose that the area of $\triangle DPQ$ is 10. Find the area of quadrilateral $ABDC$.

Problem 9:
Ryan types the string of letters AMCAMCAMCAMCAMC. Each move, Taiki chooses a gap between two adjacent letters in the current string uniformly at random and types the letter D in that gap. After six moves, the probability that the letters A, M, C do not appear consecutively in that order is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the remainder when $m + n$ is divided by 1000.

Problem 10:
For a positive integer n, let $s(n)$ denote the sum of the digits of the base-3 representation of n. Let N be the number of positive integers n less than 3^{12} such that

$$s(757n) = 3s(n).$$

Find the sum of the distinct prime divisors of N.

Problem 11:
Let x, y, and z be (not necessarily distinct) nonzero complex numbers such that

$$x^2 + y^2 + z^2 = x + y + z = x^{-1} + y^{-1} + z^{-1} \quad \text{and} \quad x^3 + y^3 + z^3 = 300.$$

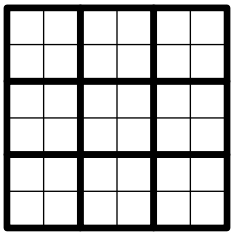
Find the sum of all possible values of $(x + 1)(y + 1)(z + 1)$.

Problem 12:
In $\triangle ABC$ with $\angle B = 72°$, $\angle C = 48°$, and incenter I, let E and F lie on $\overline{AB}$ and $\overline{AC}$, respectively, so that E, F, and I are collinear. Suppose $BE + CF = EF$ and $AE > AF$. Find the degree measure of $\angle AIE$.

Problem 13:
Let m and n be positive integers. There are exactly 14 ordered pairs of real numbers (s, t) with $0 < s, t < 1$ such that $ms + 3t$ and $2s + nt$ are both integers. Find the sum of all possible values of $2m + 3n$.

Problem 14:
Points $P_1(0, 12)$, $P_2(-5, 0)$, and $P_3(9, 0)$ lie in the xy-plane. Point $A(a, b)$, distinct from P_1, P_2, and P_3, lies in the xy-plane such that for all integers i, line AP_i passes through the circumcenter of $\triangle AP_{i+1}P_{i+2}$, where $P_{i+3} = P_i$ for all i. Find the sum of all possible values of $a \cdot b$.

Problem 15:
In the 6×6 grid of cells shown below, let N be the number of ways to place a check mark in exactly one cell of each of the nine heavily outlined 2×2 squares of cells such that for each of the twenty-five 2×2 squares of cells in the grid, at least one of its cells contains a check mark.



Find the remainder when N is divided by 1000.